Numerical Methods

Section 1: Location of Roots & Sign Change

1 Show that the equation $x^3 - 2x - 5 = 0$ has a real root in the interval $[2, 3]$.

a. $f(2) = -1 < 0$ and $f(3) = 16 > 0$, so by IVT a root exists

b. $f(2) = 1 > 0$ and $f(3) = -2 < 0$

c. $f'(x) > 0$ for all x

d. $f(2.5) = 0$

2 Under which condition does a sign change of $f(x)$ between a and b fail to guarantee a root in (a, b)?

a. When $f(x)$ is not continuous on $[a, b]$

b. When $f'(x)$ changes sign

c. When $f(x)$ has an inflection point

d. When the interval is wider than 1 unit

3 Use the sign change method to determine which interval of length 0.1 contains the root of $x^3 = 20$.

a. $[2.7, 2.8]$

b. $[2.5, 2.6]$

c. $[2.6, 2.7]$

d. $[2.8, 2.9]$

Section 2: Fixed-Point Iteration $x_{n+1} = g(x_n)$

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4 Rearrange the equation $x^3 - x - 1 = 0$ into an iterative formula $x_{n+1} = g(x_n)$.

a. $x_{n+1} = \sqrt[3]{x_n + 1}$

b. $x_{n+1} = x_n^3 - 1$

c. $x_{n+1} = \dfrac{1}{x_n^2 - 1}$

d. $x_{n+1} = x_n^3 + x_n$

5 State the condition on $g'(x)$ for the iteration $x_{n+1} = g(x_n)$ to converge to a root α.

a. $|g'(\alpha)| < 1$

b. $|g'(\alpha)| > 1$

c. $g'(\alpha) = 0$

d. $g'(\alpha) > 0$

6 Starting with $x_1 = 1.3$, use $x_{n+1} = \sqrt[3]{x_n + 1}$ to calculate x_2 and x_3 correct to 3 decimal places.

a. $x_2 = 1.320, x_3 = 1.324$

b. $x_2 = 1.300, x_3 = 1.310$

c. $x_2 = 1.350, x_3 = 1.380$

d. $x_2 = 1.315, x_3 = 1.330$

Section 3: Cobweb and Staircase Diagrams

7 If $0 < g'(\alpha) < 1$ at the root, what pattern is produced on the iteration diagram?

a. A staircase diagram converging to the root

b. A cobweb diagram converging to the root

c. A diverging staircase diagram

d. An oscillating cycle

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8 If $-1 < g'(\alpha) < 0$ at the root, what pattern is produced on the iteration diagram?

a. A cobweb diagram spiraling inwards (converging)

b. A staircase diagram spiraling outwards

c. A cobweb diagram spiraling outwards (diverging)

d. A vertical asymptote

9 What happens when $|g'(\alpha)| > 1$?

a. The iteration diverges away from the root

b. The iteration converges very quickly

c. The iteration oscillates between two fixed values

d. The root becomes imaginary

Section 4: Newton-Raphson & Advanced Applications

10 State the Newton-Raphson iteration formula for finding roots of $f(x) = 0$.

a. $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$

b. $x_{n+1} = x_n + \dfrac{f(x_n)}{f'(x_n)}$

c. $x_{n+1} = x_n - \dfrac{f'(x_n)}{f(x_n)}$

d. $x_{n+1} = \dfrac{x_n f'(x_n) - f(x_n)}{2}$

11 Use one iteration of Newton-Raphson with $x_0 = 2$ to approximate the root of $f(x) = x^2 - 5$.

a. $x_1 = 2.25$

b. $x_1 = 2.20$

c. $x_1 = 2.30$

d. $x_1 = 2.15$

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12 Under which circumstance will the Newton-Raphson method fail completely at an iteration step x_n?

a. When $f'(x_n) = 0$ (tangent is horizontal)

b. When $f(x_n) = 0$

c. When $f''(x_n) = 0$

d. When x_n is negative

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